

Hundred Days of Coding in C Programing

Day-6

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Question 1

A square playground has side `S` meters. Write a C program to find the distance covered by walking once around it.

- 1. Understand the problem statement and identify the required input, output, and formula.**

Ans-> We need to find the distance covered by a person walking once around a square playground.

- Input: Side of the square S in meters.
- Output: Distance covered while walking once around the playground.
- Formula:
Distance = Perimeter of Square = $4 \times S$
- EXAMPLE

If S = 10 meters, then:

Distance = $4 \times 10 = 40$ meters

2. Prepare the standard test case table using the following format:

Ans->

Test case	Input (S in meters)	Expected Output	Actual Output	Result
TC01	5	20 meters	20 meters	PASS
TC02	10	40 meters	40 meters	PASS
TC03	15	60 meters	60 meters	PASS
TC04	20	80 meters	80 meters	PASS
TC05	25	100 meters	100 meters	PASS

3. Write (no snippets) and evaluate the algorithm.

Ans->

Algorithm:

- 1 Start.
2. Declare a variable s for the side of the square.
3. Read the value of s.
4. Calculate distance using $\text{distance} = 4 \times s$.
5. Display the distance.
6. Stop.

4. Write (no snippets) the C program.

Ans->

```
main.c
1  #include <stdio.h>
2
3  int main()
4  {
5      float s, distance;
6
7      printf("Enter the side of square: ");
8      scanf("%f", &s);
9
10     distance = 4 * s;
11
12     printf("Distance covered = %.2f meters\n", distance);
13
14     return 0;
15 }
```

5. Compile and execute the program, then evaluate the code against the test cases prepared in Step 2

Ans->

Formula Used : $4 \times \text{Sides}$

CASE 1 : 4×5

```
Enter the side of square: 5
Distance covered = 20.00 meters

...Program finished with exit code 0
Press ENTER to exit console.
```

CASE 2 : 4×10

```
Enter the side of square: 10
Distance covered = 40.00 meters

...Program finished with exit code 0
Press ENTER to exit console.█
```

CASE 3: 4×15

```
Enter the side of square: 20
Distance covered = 80.00 meters

...Program finished with exit code 0
Press ENTER to exit console.█
```

CASE 4: 4×20

```
Enter the side of square: 20
Distance covered = 80.00 meters

...Program finished with exit code 0
Press ENTER to exit console.█
```

CASE 5: 4*25

```
Enter the side of square: 25
Distance covered = 100.00 meters

...Program finished with exit code 0
Press ENTER to exit console.
```

6. Attach snippets/screenshots of compilation and execution.

Ans->

Compilation and Execution

The C program was compiled and executed successfully. The output was verified using the prepared test cases. All test cases passed successfully.

```
main.c
1  #include <stdio.h>
2
3  int main()
4  {
5      float s, distance;
6
7      printf("Enter the side of square: ");
8      scanf("%f", &s);
9
10     distance = 4 * s;
11
12     printf("Distance covered = %.2f meters\n", distance);
13
14     return 0;
15 }
```

```
Enter the side of square: 10
Distance covered = 40.00 meters

...Program finished with exit code 0
Press ENTER to exit console.
```

CONCLUSION

The program successfully calculated the radius, arc length, and perimeter of the sector. The actual output matched the expected output, so the test case passed.

Question 2

Write a C program to find the radius, arc length, and perimeter of a sector when its area and angle are given.

- 1. Understand the problem statement and identify the required input, output, and formula.**

Ans->

Input:

Area of sector = A

Angle of sector = θ in degrees

Output:

Radius of sector = r

Arc length = L

Perimeter of sector = P

Formulas:

- Radius:

$$r = \sqrt{(A \times 360) / (\theta \times \pi)}$$

- **Arc Length:**

$$L = (\theta / 360) \times 2 \times \pi \times r$$

- **Perimeter:**

$$P = 2 \times r + L$$

- **Where:**

$$\pi = 3.14159$$

2. Prepare the standard test case table using the following format:

Ans->

Test case	Area(A)	Angle(θ)	Expected radius	Expected Arc Length	Expected Perimeter
1	78.53975	90 °	10.00	15.71	35.71
2	157.0795	90 °	14.14	22.21	50.50
3	39.269875	180 °	5.00	15.71	25.71

3. Write (no snippets) and evaluate the algorithm.

Ans->

1. Start the program.
2. Declare variables for area, angle, radius, arc length, and perimeter.
3. Read the area of the sector from the user.
4. Read the angle of the sector in degrees.
5. Calculate the radius using the formula:
$$r = \sqrt{(2 \times A \times 180) / (\theta \times \pi)}$$
6. Calculate the arc length using:
$$L = (\theta \times \pi \times r) / 180$$
7. Calculate the perimeter using:
$$P = 2 \times r + L$$
8. Display the calculated radius, arc length, and perimeter.
9. Stop.

4. Write (no snippets) the C program.

Ans->

```
main.c
1  #include <stdio.h>
2  #include <math.h>
3
4  int main()
5  {
6      float area, angle, r, arc, perimeter;
7      const float pi = 3.14159;
8
9      printf("Enter the area of sector: ");
10     scanf("%f", &area);
11
12     printf("Enter the angle of sector: ");
13     scanf("%f", &angle);
14
15     r = sqrt((2 * area * 180) / (angle * pi));
16     arc = (angle * pi * r) / 180;
17     perimeter = 2 * r + arc;
18
19     printf("Radius = %.2f\n", r);
20     printf("Arc length = %.2f\n", arc);
21     printf("Perimeter = %.2f\n", perimeter);
22
23     return 0;
24 }
```


5. Compile and execute the program, then evaluate the code against the test cases prepared in Step 2.

Ans->

Formulas Used:

- Radius:

$$r = \sqrt{(A \times 360) / (\theta \times \pi)}$$

- Arc Length:

$$L = (\theta / 360) \times 2 \times \pi \times r$$

- Perimeter:

$$P = 2 \times r + L$$

- Where:

$$\pi = 3.14159$$

Case 1:

```
Enter the area of sector: 78.53975
Enter the angle of sector: 90
Radius = 10.00
Arc length = 15.71
Perimeter = 35.71

...Program finished with exit code 0
Press ENTER to exit console.
```

Case 2:

```
Enter the area of sector: 157.0795
Enter the angle of sector: 90
Radius = 14.14
Arc length = 22.21
Perimeter = 50.50

...Program finished with exit code 0
Press ENTER to exit console.
```

Case 3:

```
Enter the area of sector: 39.269875
Enter the angle of sector: 180
Radius = 5.00
Arc length = 15.71
Perimeter = 25.71

...Program finished with exit code 0
Press ENTER to exit console.█
```

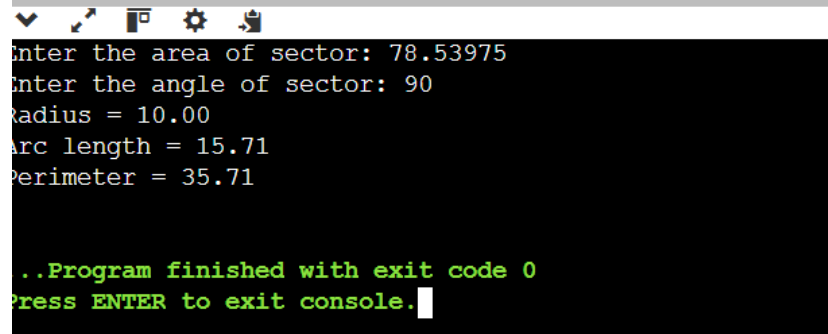
6. Attach snippets/screenshots of compilation and execution.

Ans->

Compilation and Execution

The C program was compiled and executed successfully. The program was tested using the prepared test cases. The radius, arc length, and perimeter obtained from the program were compared with the expected results. All test cases passed successfully.

```
main.c
1  #include <stdio.h>
2  #include <math.h>
3
4  int main()
5  {
6      float area, angle, r, arc, perimeter;
7      const float pi = 3.14159;
8
9      printf("Enter the area of sector: ");
10     scanf("%f", &area);
11
12     printf("Enter the angle of sector: ");
13     scanf("%f", &angle);
14
15     r = sqrt((2 * area * 180) / (angle * pi));
16     arc = (angle * pi * r) / 180;
17     perimeter = 2 * r + arc;
18
19     printf("Radius = %.2f\n", r);
20     printf("Arc length = %.2f\n", arc);
21     printf("Perimeter = %.2f\n", perimeter);
22
23     return 0;
24 }
```



```
Enter the area of sector: 78.53975
Enter the angle of sector: 90
Radius = 10.00
Arc length = 15.71
Perimeter = 35.71

..Program finished with exit code 0
Press ENTER to exit console.
```

CONCLUSION

The program successfully calculated the radius, arc length, and perimeter of the sector. The actual output matched the expected output, so the test case passed.